Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2633012

Summary

Problem Analysis: We decompose total current draw into processor, network, and display subsystems, each gated by a screen-state indicator $S(t)$, and model network polling and CPU load as truncated Fourier series over a stochastic usage schedule within a 16-hour waking window.

Building the ODE Model: Starting from the Coulomb-counting equation $\dot{z}(t) = -\eta(t)\, I(t)/Q(t)$, we couple three nonlinear feedback mechanisms: a Newton-cooling thermal ODE for battery temperature $T(t)$, an Arrhenius–SEI model for Coulombic efficiency $\eta(t)$, and the Peukert effect for effective capacity $Q(t)$.

Sensitivity Analysis: A global sensitivity sweep via UQLab over all 22 parameters reveals that the processor active-draw coefficient α_2 is the dominant contributor to SOC depletion, with passive and connectivity terms playing measurably smaller roles.

Calculating Time-to-Empty (TTE): Numerical integration of the coupled system predicts time-to-empty across a range of initial charges and usage intensities, isolating which scenarios produce rapid drain versus those that change the prediction surprisingly little.

Conclusions and Recommendations: Disabling processor-heavy background tasks and enabling adaptive brightness yield the largest marginal gains in battery life; we discuss how the framework could extend to long-term cycle-life degradation and other portable devices.

By treating battery depletion as a continuous process governed by coupled electrochemical and thermal dynamics rather than a linear subtraction of charge, our framework offers a physics-based tool for understanding and mitigating "battery anxiety."

Keywords: State of Charge; Coulomb Counting; Peukert Effect; Thermal Coupling; Global Sensitivity Analysis; Time-to-Empty

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1 Introduction

Smartphones have become an indispensable tool in the modern digital age across nearly ever facet of society. Even as chip architecture and processing power grow exponentially, the utility of these mobile devices remains tethered to the laws of thermodynamics and electrochemistry; specifically, practically all consumer electronics rely on lithium-ion batteries. Even as battery capacities grow with each new generation of electronics, battery depletion still remains a frustrating, confusing, and seemingly inconsistent process among consumers.

In reality, battery depletion is a complex process involving power consumption from various active and passive processes as well as environmental factors. The perceived inconsistencies in battery life can thus be attributed to coupled, non-linear systems interacting across multiple scales. Problem A asks us to create a continuous-time model that encompasses these dynamic processes to predict time-to-empty (TTE) under typical usage conditions. Our ODE model will capture these complex interplays to analyze how the state of charge (SOC) changes over the course of a day.

This report provides a rigorous framework for developing our ODE model. We begin with a simple depiction of battery depletion in Section 1 and extend this idea to encompass other factors that can influence SOC and TTE including user behavior, background processes, processor load, and environmental conditions in Section 2. We then validate our model by correlating parameter values with existing research and performing sensitivity analysis in Section 3. Section 4 focuses on calculating TTE under various conditions, and Section 5 concludes our report with recommendations and other findings.

2 Establishing a Baseline Time-Continuous ODE

2.1 Background

To develop a framework grounded in physics and chemistry, we first began with a simple model of battery depletion. This formulation represents the most intuitive physical interpretation of a battery: a chemical reservoir governed by the Law of Conservation of Charge. Previous work in the field has established the Coulomb Counting method as a standard for estimating SOC as a function of time (Shuzhi et al., 2021).

$$SOC(t) = SOC(0) + \frac{1}{Q_{capacity}}(\int_0^t \eta_i I(t)dt) \tag{1}$$

While the integral form does provide a cumulative sum of charge, we differentiate the model to obtain an explicit continuous-time Ordinary Differential Equation (ODE). This shift satisfies the core requirement of Problem A and allows us to later characterize of battery depletion as a dynamic, evolving state.

$$\dot{z}(t) = -\frac{\eta_i I(t)}{Q_{capacity}} \tag{2}$$

This model defines $Q_{capacity}$ as the capacity of the lithium ion battery, η_i as the Coulombic Efficiency, and $I(t)$ as the total current drawn by phone processes. We chose this Coulombic framework as the "skeleton" of our model for three reasons:

- **Physical Interpretability:** This framework allows us to ground our model in line with

electrochemistry, rather than creating a "black box" system. This phenomenological approach can be more resource-intensive to model, but it allows us to more accurately capture the thermodynamic processes involved in battery depletion. For instance, the negative sign represents the movement of lithium ions from the anode, which directly reduces the stored chemical potential of the cell during discharge and effectively represents a reduction in SOC.

- **Dimensional Normalization:** Dividing the current (mA) by the capacity (mAh), the resulting derivative $\dot{z}(t)$ is expressed in units of h^{-1}. This ensures that our state variable $z(t)$ remains strictly normalized between 1.0 (100%) and 0.0 (0%), which is consistent with the battery levels shown on smartphones. This allows us to vary variables and parameters as desired while maintaining functionality.
- **Modular Extensibility:** This ODE allows for modularity. For instance, it provides the mathematical structure necessary to decompose the total current $I(t)$ into specific hardware functions. We use this feature in Section 3 to account for processes such as display luminance and processor load.

By establishing this basis for our model, we create a strong baseline against which we can account for more complicated user behaviors, such as multiple short sessions or intensive, long sessions. Additionally, this framework gives us the freedom to incorporate the effects of physical phenomena such as Solid-Electrolyte Interphase (SEI) and Peukert Effect.

3 Non-Linear Coupling for Dynamic State-of-Charge Declination

3.1 User Behaviors

This section details the physical rationale behind our governing equations as they relate to $\dot{z}(t)$. Defining the effects of various active and passive processes allows us to capture the effects of dynamic user and environmental changes on SOC. We will first characterize $I_{total}(t)$ by breaking it down into its constitutive functions.

Over the course of a day, active user actions such as screen brightness, network communications, and application usage have the largest impacts on how smartphone battery declines throughout the day. In our framework, the fundamental gate for these power-intensive processes is the screen state, or whether the screen is on or off. We utilize an indicator function $S(t)$ to represent this binary state:

$$S(t) = \begin{cases} 1 & t \in \{\text{using phone}\} \\ 0 & \text{else} \end{cases} \tag{3}$$

Rationale: This function serves as the "master switch" in our ODE system. All of the active user processes we define later in this section only contribute to a change in SOC when the phone is on and actively being used.

To simulate periods of active phone use, we first defined the interval during which a smartphone could be actively used as any time between 07:00 and 23:00, which is a 16-hour "Active" interval centered around 12:00. We chose this interval under the assumption that the user will be asleep, and thus be unable to use their phone, for 8 hours during a typical night. Although many adults report inconsistent levels of sleep in the modern day, this is consistent with existing sleep data

and recommendations (Scott et al., 2024). We then stochastically divided the Active interval into n_1 periods of time that sum to a total screen-on time n_2. Given that we are constructing a continuous-time model, we can arbitrarily change these values to represent any level of user activity within the Active interval. For example,
$\{t \in [10, 14] \cup [16\frac{30}{60}, 16\frac{45}{60}]\}$ would represent someone using their smartphone continuously between 10:00 and 14:00 and also for just 15 minutes between 16:30 and 16:45. Figure 1 represents a seed where $n_2 = 6$ hours where $n_2 = 15$ minutes. Note that all Figures used in this section are run on this seed as an example.

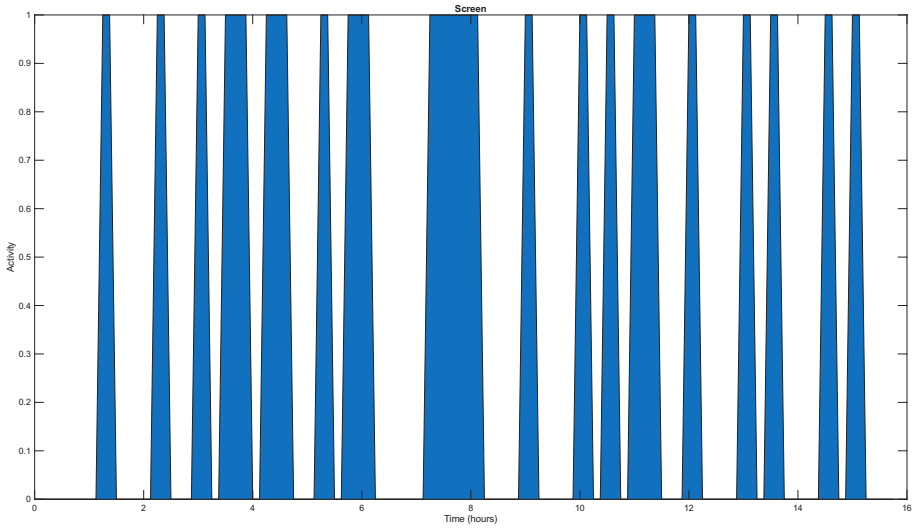


Figure 1: Example Screen Time Indicator Function

3.1.1 User Sources of Current Draw

Breaking down $I(t)$ into its constitutive functions, we get the following breakdown:

$$I(t) = I_{processor} + I_{connectivity} + I_{screen} \tag{4}$$

These constitutive functions represent three major categories of active battery draw related specifically to active user usage. It is important to note that all of these processes also have passive components, which will also be addressed in this section.

Network usage and polling involves WiFi, mobile, Bluetooth, and other network connections and communications. We model the network usage over the course of a day as a continuous periodic curve via a truncated Fourier series.

$$N(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_1 t) + b_n \sin(n\omega_1 t) \approx \gamma_1 + \gamma_2 \cos(\omega_1 t) + \gamma_3 \sin(\omega_1 t) \tag{5}$$

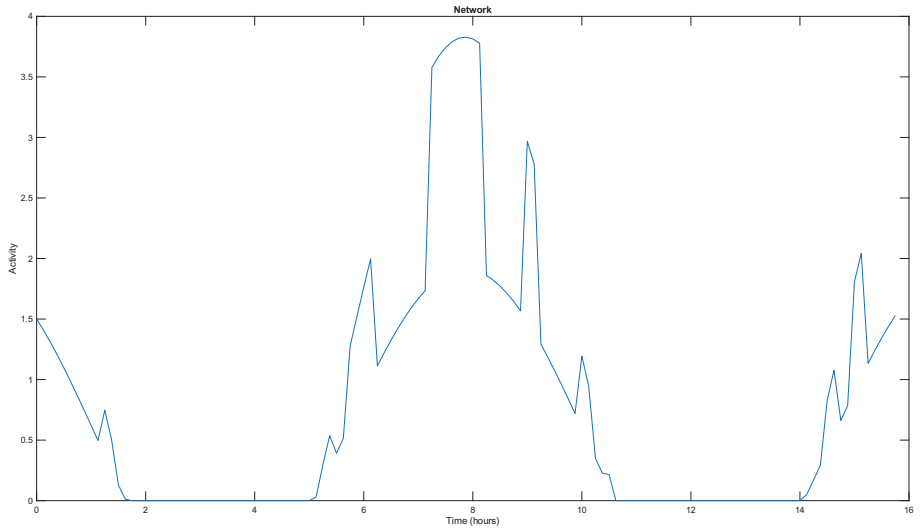


Figure 2: Example Network Usage Function

We thus define the current draw from connectivity as the following:

$$I_{connectivity}(t) = (0.5S(t) + 0.5) * \left[I_{conec,use} N(t) + I_{conec,tail} N(t - \frac{1}{60}) \right] + I_{conc,base} \tag{6}$$

where $I_{conec,base}$ is the passive current draw of network services, $I_{conec,use}$ is the average current draw when using network services, and $I_{conec,tail}$ represent the current due to the average current draw in the tail stage of using network services. This tail stage accounts for roughly 60% of battery drain caused by connection services as the device must remain connected to the network after active usage to continue its final communications. The $0.5S(t) + 0.5$ term helps scale the current draw by active device usage.

Processor load due to application usage indicates how computationally heavy a user session is. This essentially represents how the battery responds to specific apps the user uses. For instance, the battery responds differently to high-consumption gaming sessions and simple SMS texting. Similarly to network usage, we model processor load as a continuous periodic curve and represent it with a Fourier series:

$$U(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_2 t) + b_n \sin(n\omega_2 t) \approx \beta_1 + \beta_2 \cos(\omega_2 t) + \beta_3 \sin(\omega_2 t) \tag{7}$$

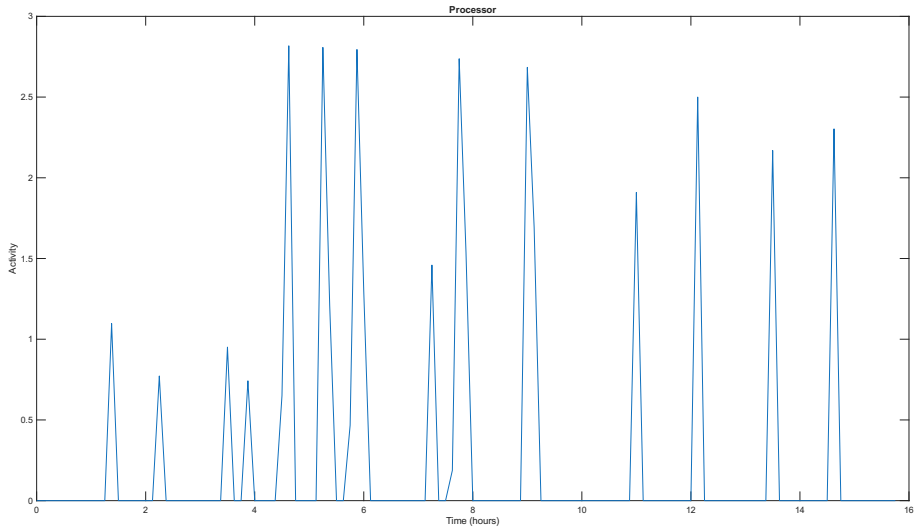


Figure 3: Example CPU Usage Function

We thus define current draw due to processor load as the following:

$$I_{processor}(t) = S(t) * I_{pros,use}U(t) + I_{pros,base} \tag{8}$$

where $I_{pros,base}$ is the passive current draw of the processor and $I_{pros,use}$ is the maximum current draw minus the baseline current draw. The $S(t)$ term once again ensures that active draw only occurs while the phone is actively being used.

Rationale: This Fourier approach for network connectivity and processor load allows the model to simulate the "waves" of activity common in modern life, such as high-intensity morning usage followed by idle periods. It also captures "spikes" in network activity that may occur when syncing data, uploading media, or making phone calls. By using a continuous periodic curve, we satisfy the requirement for a continuous-time model while allowing for parameter estimation that reflects realistic human habits.

Nearly all modern smartphones utilize adaptive brightness (Ma et al., 2012), which is a feature that automatically adjusts the display's brightness based on external light conditions. As such, we chose to define the current drawn by lighting the screen based on the thermal radiation produced by the Sun throughout the day. We fit our parameter values for a sinusoidal function $\hat{y}_2$ based on the model provided by Karim and Singh, 2014 that models this approach:

$$\hat{y}_2 = a_1 * \sin(b_1 t + c_1) + a_2 * \sin(b_2 t + c_2) \tag{9}$$

We implemented this in an equation for adaptive screen brightness $B(t)$, which we shifted and stretched to fit our Active interval.

$$B(t) = \sqrt{\hat{y}_2(t * \frac{22}{16} + 1)} \tag{10}$$

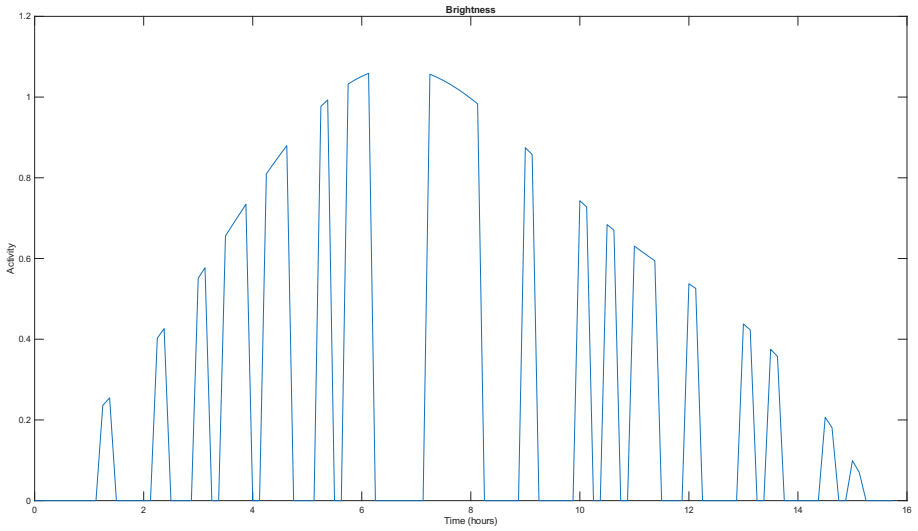


Figure 4: Example Screen Brightness Function

Thus, the current draw from brightness is as follows:

$$I_{screen}(t) = S(t)I_{screen,use}B(t) + I_{screen,base} \tag{11}$$

where $I_{screen,base}$ is the passive current draw of the electronics that control brightness (even while the display is turned off) and $I_{screen,use}$ is the average current draw of the screen. $S(t)$ scales the current draw by active device usage, as the majority of the screen draw only occurs while the screen is drawing power.

Combining all of the phenomena and functions above, we get the following $I(t)$ representing the summative effects of passive and active processes. Of note, the c term in the equation below represents the cumulative sum of all passive effects for each of these current draws.

$$I(t) = \left[\alpha_1 U(t) + \alpha_2 APL(t)B(t)\right] S(t) + \left[0.5S(t) + 0.5\right] \left[\alpha_3 N(t) + \alpha_4 N(t - \frac{1}{60})\right] + \alpha_5 \tag{4}$$

3.2 Environmental and Passive Processes

While an electronic device is in active use, it will generate heat. Most modern phones have robust thermal regulation capabilities that keep them within the normal range of 0°C and 40°C (Visuri et al., 2022). However, in extreme external environments or during long, strenuous user sessions, the temperature can move outside this interval. In regards to internal electronics, the rate of change of battery temperature is described below.

$$\dot{T} = \left[I(t)^2 R - hA(T - T_0)\right] / C \tag{12}$$

Where R is the internal resistance of the battery in Ω, hA is the heat transfer coefficient by area measured in W/K, T_0 is the ambient temperature in K, and C is the thermal mass in J/K. The actual effect temperature has on battery discharge functionality is defined by the Coloumbic Effect $\eta(T)$, which essentially represents the efficiency at which the battery is able to discharge.

$$\eta(t) = 1 - \frac{I_{loss}(t)}{I_{applied}(t)} = 1 - \frac{\eta_1 |I(t)|^{\eta_2} * e^{-E_A/RT}}{I(t)} \tag{13}$$

Where η_1 is the one minus the baseline efficiency of the battery, η_2 is the SEI loss, E_A is the activation energy of the electrons in J/mol, and R is the gas constant in $J/molK$. SEI loss is essentially a protective feature of the battery that limited efficiency but ensures functionality. In typical use cases, this effect typically remains between 0.99 and 1.00, representing little change in the overall SOC.

Another factor to consider is the effective capacity due to the Peukert Effect, where high-current draw makes a portion of the total capacity Q_{total} inaccessible. This explains why a device drains faster under heavy load than its theoretical maximum capacity would suggest.

$$Q(t) = Q_{total} \frac{1}{I(t)}^{1-k} \tag{14}$$

Here, Q_{total} is the total capacity of the battery, while k is the Peukert constant.

3.3 Assumptions

The following assumptions underpin the model and constrain its domain of validity. Each is either imposed directly by our modelling choices or inherited from the parameter values established in Sections 3.1 and 3.2.

- **Initial condition.** The battery begins each simulation cycle at full charge, $z(0) = 1$. This corresponds to a device that was fully charged overnight before the 16-hour active window begins at 07:00.
- **Active window and screen time.** The user is awake and potentially active only between 07:00 and 23:00, a 16-hour window consistent with standard sleep recommendations (Scott et al., 2024). Within this window, total screen-on time is fixed at 6–7 hours per day, in accordance with reported U.S. averages. The specific screen-on intervals are drawn stochastically via the indicator function $S(t)$; phase randomisation ensures that the results are not tied to a single usage schedule.
- **Single network connection.** At any instant the device maintains at most one active network connection—either Wi-Fi or a cellular data link, but not both simultaneously. Wi-Fi and cellular are modelled with the identical Fourier-plus-tail structure of Equation (6); only the magnitudes of the active-draw and tail-draw coefficients α_3 and α_4 differ between the two connection types.
- **OLED display with solar-driven adaptive brightness.** The display is assumed to be OLED, for which the current drawn by the screen is proportional to the instantaneous brightness level. Brightness is controlled exclusively by the adaptive-brightness algorithm described in Section 3.1.1: it follows the double-sinusoid solar-irradiance proxy $B(t)$ of Karim and Singh (2014) and is not overridden by manual user input.
- **Constant ambient temperature.** The surrounding environment is held at $T_0 = 300$ K ($\approx 27\,^{\circ}$C) throughout the simulation. This isolates the internal Joule-heating dynamics modelled by $\dot{T}$ from external weather variation.
- **No charging during the simulation window.** The device is not connected to a power source at any point during the 16-hour discharge cycle; $z(t)$ is strictly non-increasing.

- **Within-cycle electrochemistry only.** The Coulombic efficiency $\eta(t)$ and the internal resistance R are treated as functions of instantaneous current and temperature within a single discharge. Long-term cycle-life degradation (e.g. State-of-Health loss over hundreds of charge cycles) is not modelled. Under typical single-cycle conditions $\eta(t)$ remains in the narrow band [0.99, 1.00].

4 Model Analysis

4.1 Parameter Estimation

The parameters in our model fall into three categories: hardware dependent constraints, electrochemical coefficients, and behavioral usage parameters. As such, we used a combination of physics and literature to get initial estimates of the values.

4.1.1 Hardware and Electrical Parameters

The current draw coefficients, $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$, all represent physical power consumption of electronics. As such, we used publicly available analyses and profiling studies of modern smartphones to obtain our estimates. Often, this approach also yields accurate parameter ranges as well as point estimates.

The battery capacity parameters were also chosen using widely available public information. The total battery capacity was easily found online. The Peukert constant for lithium-ion cells is also a well known value in the relevant literature. The final estimates can be seen in table 1.

Current ($I(t)$) Parameters			
α_1	Screen draw (mA)	300	50–500
α_2	Processor active draw (mA)	1400	20–1800
α_3	Network active draw (mA)	800	400–1400
α_4	Network tail draw (mA)	200	50–500
α_5	Passive draw (mA)	300	200–500
Effective Capacity ($Q(t)$) Parameters			
Q_{total}	Total charge (mAh)	5000	3000–6000
$I_{reference}$	Reference current (mA)	100	–
k	Peukert Constant	1.10	1.03-1.15

Table 1: hardware and Electrical Parameter Estimates

4.1.2 ElectricalChemical Parameters

Thermal Parameters were estimated using a combination of literature and physical knowledge. Internal resistance of the battery was estimated using public information of small-format lithium-ion cells. Thermal mass and the cooling coefficient were estimated using their respective formulas and information about basic constants found online. The final estimates can be seen in Table 2.

The Coulombic Parameters were harder to estimate and required more information to estimate. There is limited information on SEI exponents specifically for smartphone batteries, so we relied on estimates from other models. Activation energy E_A was found in experimental literature, while

η_1 was chosen to best fit the model to reported experimental values. The final parameter estimates can be seen in Table 2.

Temperature ($T(t)$) Parameters			
T1	Internal resistance (Ohm)	0.08	0.05–0.10
T2	Cooling coefficient (W/K)	3	0.5–5
T3	Ambient temperature (K)	300	293–307
T4	Thermal mass (J/K)	200	150–300
Columbic Efficiency $\eta(t)$ Parameters			
I1	Baseline loss term	10^{-2}	$10^{-2} - 10^{-3}$
I2	SEI loss exponent	1.15	1.03–1.20
I3	Activation energy (J/mol)	3000	2000–5000
I4	Gas constant (J/mol K)	8.314	–

Table 2: Thermal Parameter Estimates

4.1.3 Behavioral Parameters

User behavior functions $S(t), U(t), N(t), B(t)$ were parameterized using bounded Fourier components. Coefficients were chosen to reproduce realistic daily usage variability rather than to fit a specific user. Screen time was fixed at 6–7 hours per day in accordance with U.S. averages, while phase randomization ensured that results were not dependent on a particular usage schedule.

All parameter values and plausible ranges are summarized in Table 3. Importantly, parameters were selected to ensure physical realism, numerical stability, and interpretability.

App Usage $A(t)$ Function			
β_1	Constant term	1	-1–1
β_2	Cosine amplitude	2	-2–2
β_3	Sine amplitude	-2	-2–2
ω_1	Frequency	0.7	0–1
Network Usage $N(t)$ Function			
γ_1	Constant term	0	-1–1
γ_2	Cosine amplitude	-2	-2–2
γ_3	Sine amplitude	2	-2–2
ω_2	Frequency	10	5-15
Brightness $B(t)$ Function			
a_1	Brightness coefficient	746	–
a_2	Brightness coefficient	77.42	–
b_1	Brightness coefficient	0.1484	–
b_2	Brightness coefficient	0.5492	–
c_1	Brightness coefficient	-0.07144	–
c_2	Brightness coefficient	2.635	–

Table 3: Behavioral Parameter Estimates

4.2 Sensitivity Analysis

To quantify the robustness of our model and identify the dominant contributors to battery depletion, we performed a global sensitivity analysis using the UQLab uncertainty quantification framework. Rather than perturbing parameters one at a time, we treated uncertain parameters as random variables and analyzed how their joint variability influences model outputs

4.2.1 Framework

We defined 22 uncertain input parameters spanning four categories:

1. User behavior dynamics (app and network usage)
2. Hardware current draw coefficients
3. Thermal parameters
4. Electrochemical efficiency parameters

Each parameter was modeled as an independent uniform random variable over a physically plausible range (Table 4). Uniform distributions were selected to reflect epistemic uncertainty rather than measurement noise, ensuring that no single parameter value was artificially favored.

The output of interest was the final State of Charge after 16 hours, with 6 hours of screen time. A Monte Carlo sample of 10^4 parameters realizations was generated to ensure statistical convergence of correlation estimates.

4.2.2 Results

Figure 5 presents the correlation-based sensitivity indices for all uncertain model parameters. Both linear (Pearson) and rank (Spearman) correlation coefficients are shown to capture linear and monotonic nonlinear dependencies between inputs and the final State of Charge (SOC) after 16 hours.

The analysis reveals that current draw coefficients dominate system behavior. In particular, the processor current coefficient A_2, network current coefficient A_3, and passive background draw A_5 exhibit the strongest negative correlations with final SOC. This indicates that increases in computational intensity, network usage, and baseline system load lead to consistently faster battery depletion. Among all parameters, A_2 displays the largest magnitude correlation, highlighting the dominant role of processor load in determining battery life.

Behavioral usage parameters associated with app and network activity (β_i, γ_i, ω_1, ω_2) show moderate influence. Frequency parameters in particular demonstrate stronger rank correlations than linear correlations, suggesting that higher-frequency usage patterns increase the occurrence of peak current events, which disproportionately accelerate SOC decay through nonlinear capacity loss mechanisms.

Electrochemical efficiency parameters governing Coulombic efficiency (I_1, I_2, and I_3) exhibit smaller but non-negligible negative correlations with SOC. These parameters do not dominate short-term depletion but become more influential under high-current and elevated-temperature regimes due to their exponential dependence on temperature and current. The stronger rank correlations observed for these parameters confirm the nonlinear nature of their effect.

Thermal parameters (T_1–T_4) show weak direct correlation with SOC. This indicates that temperature affects battery depletion primarily through indirect coupling by modifying resistance and efficiency rather than directly consuming charge. This result is consistent with the physical behavior of lithium-ion batteries.

Finally, parameters associated with screen brightness and display scaling (A_1) exhibit relatively small correlations, indicating that brightness adjustments alone have limited impact compared to processor and network loads.

Overall, the agreement between linear and rank correlation indices for dominant parameters demonstrates that the model is structurally robust and not driven by pathological nonlinearities. The results confirm that battery depletion is primarily governed by high-power subsystems and sustained current demand rather than surface-level user settings or ambient conditions.

5 Calculating Time-To-Empty

Time-To-Empty (TTE) is defined as the time at which the battery State of Charge (SOC) reaches zero under a given usage pattern. Unlike linear extrapolation methods commonly used in consumer devices, our model computes TTE by numerically integrating a coupled system of nonlinear ordinary differential equations that account for current draw, thermal effects, Coulombic efficiency, and effective capacity.

5.1 Usage Profile and Current Demand

Figures 6 and 7 illustrate representative screen activity and usage-driven demand over a 16-hour period. Screen usage is modeled as an indicator function, while processor and network activity

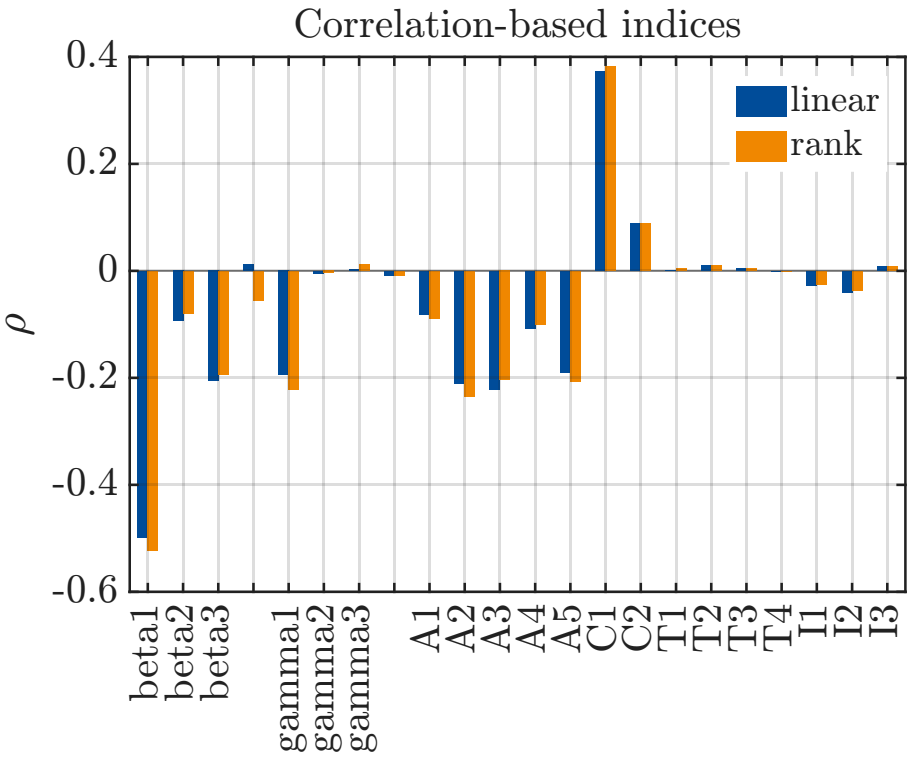


Figure 5: Sensitivity Analysis

introduce time-varying current demand. These usage patterns determine the instantaneous current $I(t)$ drawn from the battery.

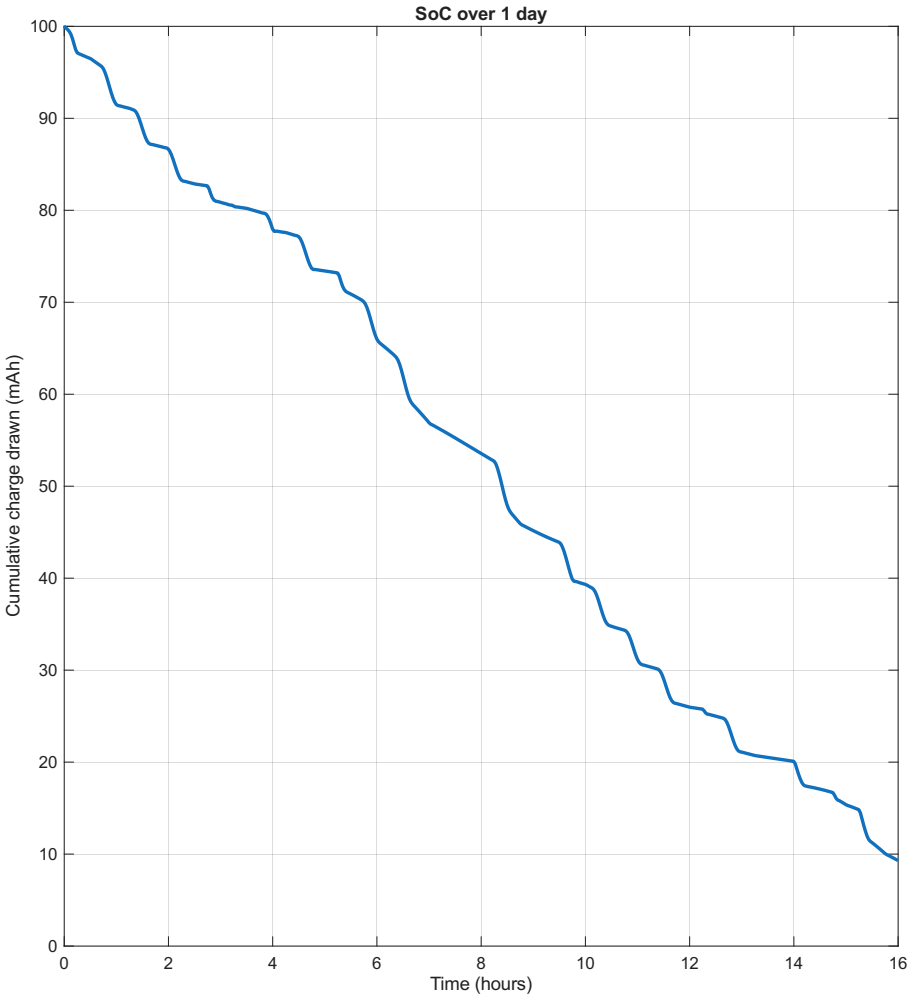


Figure 6: Screen time indicator function $S(t)$ over a 16-hour period.

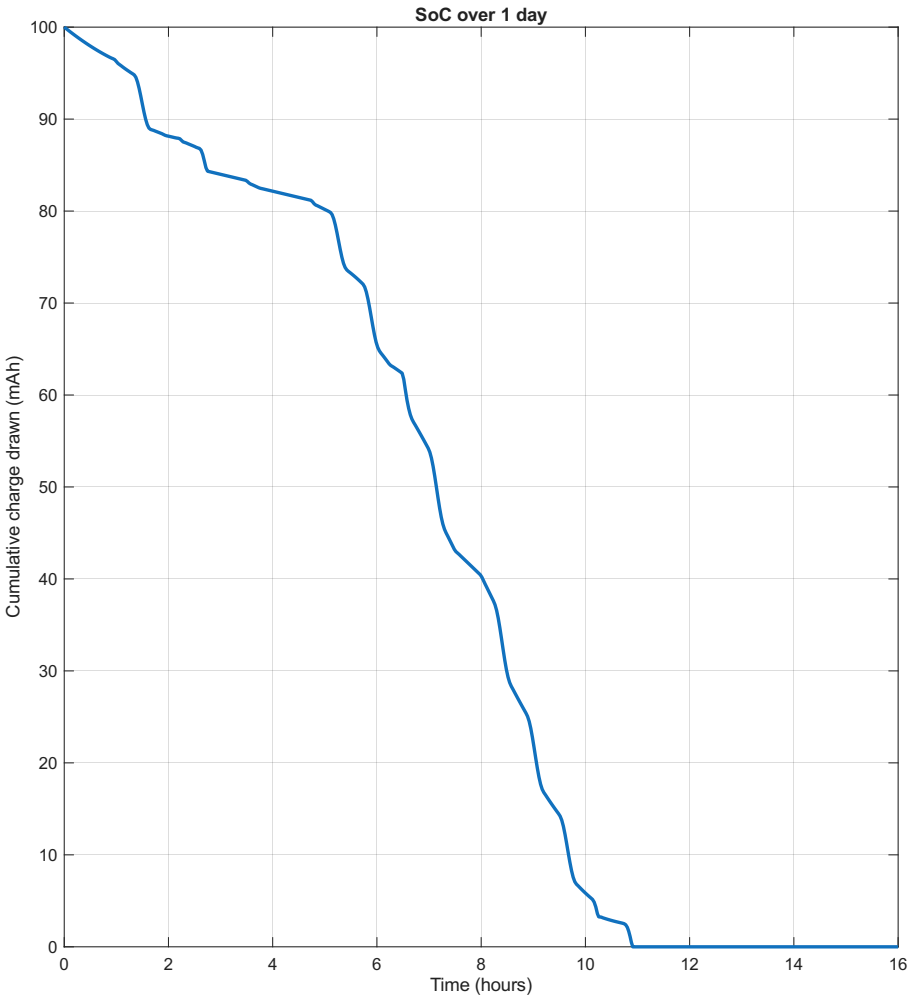


Figure 7: Cumulative charge draw and resulting State of Charge over one day.

5.2 State of Charge Evolution

The resulting SOC trajectory is shown in Figure 8. Rather than decreasing linearly, SOC exhibits curvature caused by nonlinear coupling between current draw, temperature, Coulombic efficiency, and effective capacity. Periods of sustained high usage lead to visibly steeper declines in SOC, demonstrating the compounding effect of high current demand.

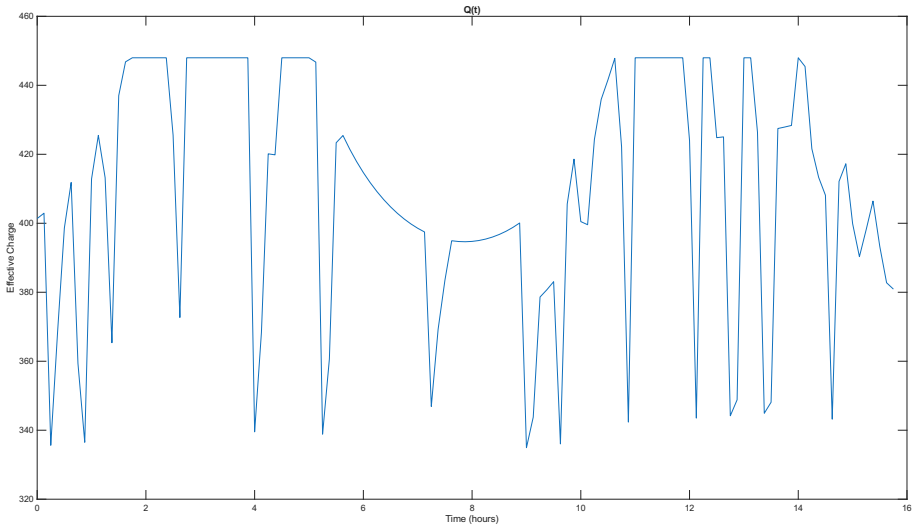


Figure 8: State of Charge (SOC) evolution over time under representative usage.

5.3 Efficiency and Thermal Effects

Figures 9 and 10 show the evolution of Coulombic efficiency and current draw, respectively. Coulombic efficiency decreases during high-current intervals due to temperature-dependent electrochemical losses, amplifying the rate of charge depletion.

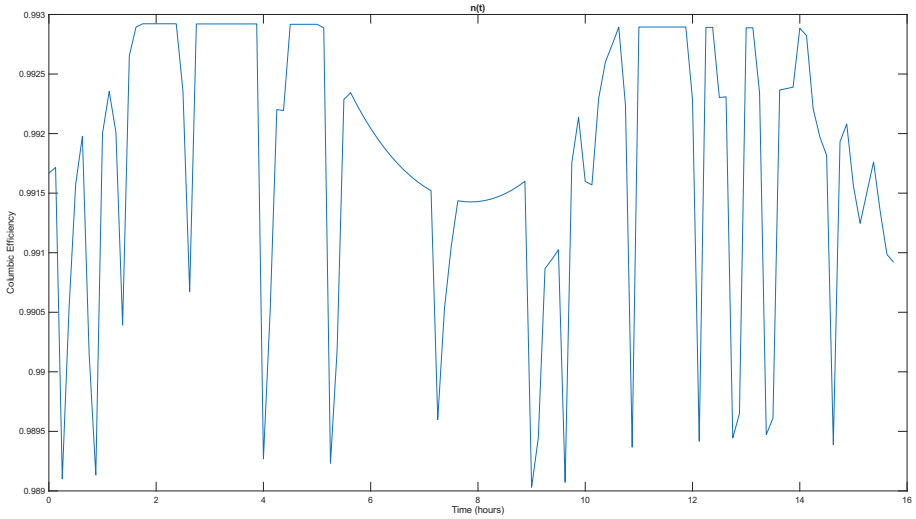


Figure 9: Coulombic efficiency $\eta(t)$ as a function of time.

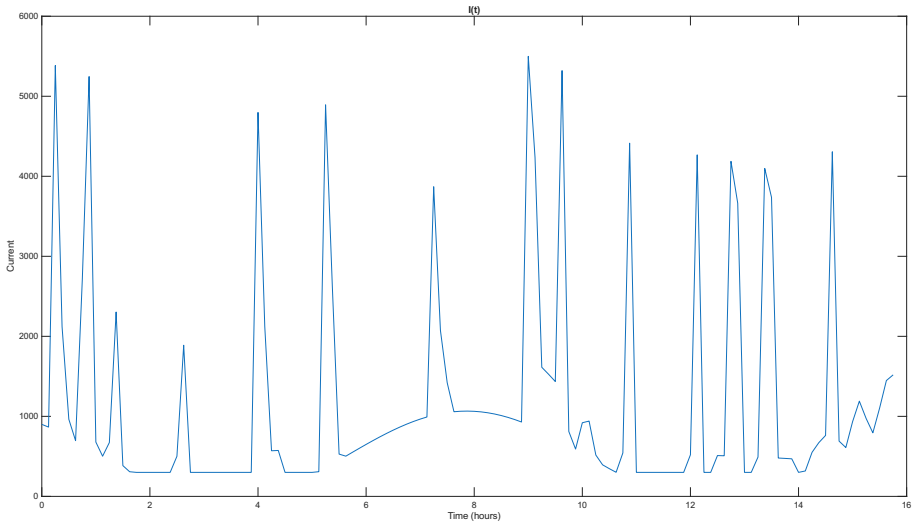


Figure 10: Instantaneous current draw $I(t)$ driven by usage patterns.

Battery temperature, shown in Figure 11, rises during periods of high current draw and decays during idle intervals. Although temperature does not directly remove charge, it indirectly accelerates depletion by increasing internal resistance and reducing efficiency.

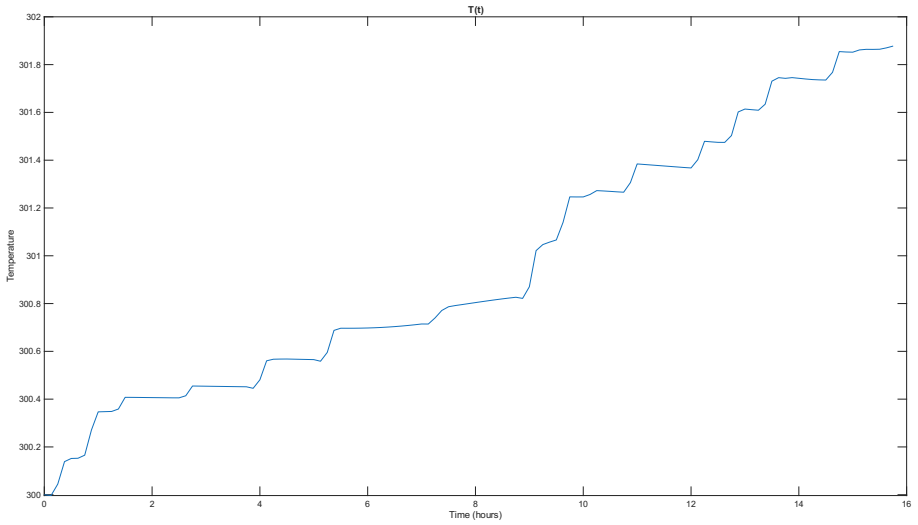


Figure 11: Battery temperature evolution resulting from resistive heating and cooling.

5.4 Determination of Time-To-Empty

Time-To-Empty is computed numerically as the first time t_{TTE} such that

$$z(t_{\mathrm{TTE}}) = 0,$$

where $z(t)$ denotes normalized SOC. The solver terminates automatically when this condition is met. Because $Q(t)$ and $\eta(t)$ depend on current and temperature, TTE varies significantly with usage intensity and timing.

This approach captures real-world phenomena such as sudden battery depletion during short periods of intensive use and explains why devices may shut down earlier than predicted by linear SOC indicators.

6 Conclusions

6.1 Key Insights from the Model

The sensitivity analysis conducted in Section 4.2 confirms that power consumption is truly a coupled response to hardware stresses. The processor current coefficient (α_2) displayed a strong negative correlation with final SOC. This implies that CPU load is a major portion of battery usage. Additionally, the tail stage power draw during network connectivity ensures that even intermittent network activity can keep the device in a high-power state. Interestingly, screen brightness (α_1) has a relatively small impact on total depletion compared to other suspected high-usage processes.

6.2 Practical Recommendations

Based on the findings of our model, here are our primary recommendations to the user to prevent "battery anxiety" and extend TTE:

1. **Manage Computational Workload:** Closing high-intensity background apps (e.g., GPS-heavy or gaming apps), limiting their usage, and reducing session times is more effective than simply lowering brightness.

2. **Optimize Connectivity:** To minimize the impact of the "tail effect," users should try to maintain stable Wi-Fi connections. Avoid changing Wi-Fi connections often and avoid switching between Wi-Fi and mobile data often. Active usage of network communications does not actually spike current draw very highly; instead, the major draw of network connectivity is the background communications that usually occur.

6.3 Future Work

Our model does not account for charging behaviors or smart battery services that affect the level the battery charges to usually. Accounting for this would allow for the model to better characterize more niche user behaviors. Additionally, testing parameters that are fit to specific devices more broadly could help generalize our findings to not just smartphones but other smart devices.

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Appendices

Appendix A

```
function Y = batteryModel_UQ(X)
% X : [N x d] matrix of inputs
% Y : [N x 1] model output (QoI)

Nsim = size(X,1);
Y = zeros(Nsim,1);

% ++++++++++++++++++ Parameters ++++++++++++++++++

% current draws of Display, Processor, Network, and Background.
A1 = 300; % c_slope, draw of pixels 50-500 (mA)
A2 = 1500; % coefficient of processor draw 0-1800 (mA)
A3 = 200; % coefficient of network draw 40-400 (wifi) 100-800 (4G) 200-1400 (5G) (
    ↪ mA)
A4 = 200; % coefficient of network tail draw 50-500 (mA)
A5 = 300; % passive draw 200+55 (base screen + base network) 200-500 (mA)

% Battery capacity
C1 = 500; % total charge (mAh)
C2 =  100;% reference current (mA)

% Temperature
T1 = 0.08; % Internal Resistance (Ohm) 0.05-0.010
T2 = 3; % Cooling Effect (W/K) 0.5-5
T3 = 300; % Ambient Temperature (K)
T4 = 200; % Thermal Mass 150-300 (J/K)
```

```
% Columbic Efficiency
I1 = 1e-2; % 1 - Baseline efficiency
I2 = 1.15; % SEI loss (diffusion limited)
I3 = 3000; % Activation Energy (J/mol)
I4 = 8.314; % Gas Constant (J/mol K)

% brightness function
a_1 = 746;
a_2 = 77.42;
b_1 = 0.1484;
b_2 = 0.5492;
c_1 = -0.07144;
c_2 = 2.635;

% time
time.end = 16;
time.step = 15/60;
time.screen = 6;
ind = randperm(time.end/time.step);

parfor i = 1:Nsim

    % -------- Unpack inputs --------

    beta1 = X(i,1);
    beta2 = X(i,2);
    beta3 = X(i,3);
    omega_1 = X(i,4);

    gamma1 = X(i,5);
    gamma2 = X(i,6);
    gamma3 = X(i,7);
    omega_2 = X(i,8);

    A1 = X(i,9);
    A2 = X(i,10);
    A3 = X(i,11);
    A4 = X(i,12);
    A5 = X(i,13);

    C1 = X(i,14);
    C2 = X(i,15);

    T1 = X(i,16);
    T2 = X(i,17);
    T3 = X(i,18);
    T4 = X(i,19);

    I1 = X(i,20);
    I2 = X(i,21);
    I3 = X(i,22);

    % varying screen time function
    S_hrs = [ones(1,time.screen/time.step),zeros(1,(time.end-time.screen)/time.step
        ↪ )];
    S_hrs = S_hrs(ind);
```

```
    % ++++++++++++++++++ Usage set over 1 day ++++++++++++++++++

    % Screen time
    S = @(t) S_hrs(min(floor(t./time.step) + 1, time.end/time.step));

    % Network Usages (call start and end times)
    N = @(t) (0.5*S(t)+0.5).*max((beta1+beta2*cos(omega_1*t)+beta3*sin(omega_1*t))
       ↪ ,0);
    N_delayed = @(t) N(max(t-1/60,0));

    % Processor Usage (based on Apps)
    U = @(t) S(t).*max(gamma1+gamma2*cos(omega_2*t)+gamma3*sin(omega_2*t),0);

    % Brightness (brightness % and time when changed)
    B = @(t) S(t).*max((a_1*sin(b_1*(t*22/16+1)+c_1)+a_2*sin(b_2*(t*22/16+1)+c_2))
       ↪ ,0)/756.8;

    % APL

    % ++++++++++++++++++ Current Draw ++++++++++++++++++
    I = @(t,I) A1*sqrt(B(t)) + A2*U(t) + A3*N(t) + A4*N_delayed(t) + A5;

    % ++++++++++++++++++ Temperature ++++++++++++++++++
    T_dot = @(t,T) ( (I(t)/100).^2*T1 - T2*(T-T3) )./T4;

    [tt, Tvals] = ode45(T_dot, [0 time.end], 300, odeset('MaxStep', time.step));

    T = @(t) interp1(tt, Tvals, t, 'linear', 'extrap');

    % ++++++++++++++++++ Columbic Efficiency ++++++++++++++++++
    nu = @(t) 1 - I1*abs(I(t)).^(I2-1).*exp(-I3./(I4*T(t)));

    % ++++++++++++++++++ Effective Capacity ++++++++++++++++++
    Q = @(t) C1.*(C2./I(t)).^(1.10-1);

    % ++++++++++++++++++ SOC Derivative ++++++++++++++++++
    z_dot = @(t,Z) -I(t)./(Q(t).*nu(t));

    % ++++++++++++++++++ Solver ++++++++++++++++++

    tspan = [0,time.end];
    x0 = 100; % starting at no charge drawn.

    opts = odeset('MaxStep', time.step/2);
    [~,Z] = ode45(z_dot, tspan, x0, opts);

    % -------- Quantity of Interest --------
    % Total charge drawn after 16 hours
    Y(i) = Z(end);

end

end
```

```
clc
```

```
clear variables
close all
clearvars
uqlab


% ++++++++++++++++++ Model ++++++++++++++++++

ModelOpts.mFile = 'batteryModel_UQ';

myModel = uq_createModel(ModelOpts);


% ++++++++++++++++++ Parameters ++++++++++++++++++

% App Usage
InputOpts.Marginals(1).Name = 'beta1'; % amplitude
InputOpts.Marginals(1).Type = 'Uniform';
InputOpts.Marginals(1).Parameters = [-1,1];

InputOpts.Marginals(2).Name = 'beta2'; % period
InputOpts.Marginals(2).Type = 'Uniform';
InputOpts.Marginals(2).Parameters = [-2,2];

InputOpts.Marginals(3).Name = 'beta3'; % x shift
InputOpts.Marginals(3).Type = 'Uniform';
InputOpts.Marginals(3).Parameters = [-2,2];

InputOpts.Marginals(4).Name = 'omega_1'; % y shift
InputOpts.Marginals(4).Type = 'Uniform';
InputOpts.Marginals(4).Parameters = [0,1];

% Network Usage
InputOpts.Marginals(5).Name = 'gamma1'; % amplitude
InputOpts.Marginals(5).Type = 'Uniform';
InputOpts.Marginals(5).Parameters = [-1,1];

InputOpts.Marginals(6).Name = 'gamma2'; % period
InputOpts.Marginals(6).Type = 'Uniform';
InputOpts.Marginals(6).Parameters = [-2,2];

InputOpts.Marginals(7).Name = 'gamma3'; % x shift
InputOpts.Marginals(7).Type = 'Uniform';
InputOpts.Marginals(7).Parameters = [-2,2];

InputOpts.Marginals(8).Name = 'omega_2'; % y shift
InputOpts.Marginals(8).Type = 'Uniform';
InputOpts.Marginals(8).Parameters = [5,15];

% Current Parameter
InputOpts.Marginals(9).Name = 'A1'; % amplitude
InputOpts.Marginals(9).Type = 'Uniform';
InputOpts.Marginals(9).Parameters = [50,500];

InputOpts.Marginals(10).Name = 'A2'; % period
InputOpts.Marginals(10).Type = 'Uniform';
```

```
InputOpts.Marginals(10).Parameters = [20,1800];

InputOpts.Marginals(11).Name = 'A3'; % x shift
InputOpts.Marginals(11).Type = 'Uniform';
InputOpts.Marginals(11).Parameters = [400,1400];

InputOpts.Marginals(12).Name = 'A4'; % y shift
InputOpts.Marginals(12).Type = 'Uniform';
InputOpts.Marginals(12).Parameters = [50,500];

InputOpts.Marginals(13).Name = 'A5'; % y shift
InputOpts.Marginals(13).Type = 'Uniform';
InputOpts.Marginals(13).Parameters = [200,500];

% Current Parameter
InputOpts.Marginals(14).Name = 'C1'; % amplitude
InputOpts.Marginals(14).Type = 'Uniform';
InputOpts.Marginals(14).Parameters = [300,600];

InputOpts.Marginals(15).Name = 'C2'; % period
InputOpts.Marginals(15).Type = 'Uniform';
InputOpts.Marginals(15).Parameters = [50,200];

InputOpts.Marginals(16).Name = 'T1'; % x shift
InputOpts.Marginals(16).Type = 'Uniform';
InputOpts.Marginals(16).Parameters = [0.05,0.15];

InputOpts.Marginals(17).Name = 'T2'; % y shift
InputOpts.Marginals(17).Type = 'Uniform';
InputOpts.Marginals(17).Parameters = [0.5,5];

InputOpts.Marginals(18).Name = 'T3'; % y shift
InputOpts.Marginals(18).Type = 'Uniform';
InputOpts.Marginals(18).Parameters = [293,307];

InputOpts.Marginals(19).Name = 'T4'; % amplitude
InputOpts.Marginals(19).Type = 'Uniform';
InputOpts.Marginals(19).Parameters = [150,300];

InputOpts.Marginals(20).Name = 'I1'; % period
InputOpts.Marginals(20).Type = 'Uniform';
InputOpts.Marginals(20).Parameters = [1e-3,1e-1];

InputOpts.Marginals(21).Name = 'I2'; % x shift
InputOpts.Marginals(21).Type = 'Uniform';
InputOpts.Marginals(21).Parameters = [1.03,1.20];

InputOpts.Marginals(22).Name = 'I3'; % y shift
InputOpts.Marginals(22).Type = 'Uniform';
InputOpts.Marginals(22).Parameters = [2000,5000];

% ++++++++++++++++++ Inputs ++++++++++++++++++

myInput = uq_createInput(InputOpts);
```

```
% ++++++++++++++++++ Analysis ++++++++++++++++++

CorrSensOpts.Type = 'Sensitivity';
CorrSensOpts.Method = 'Correlation';
CorrSensOpts.Model = myModel; % <--- add this
CorrSensOpts.Input = myInput; % <--- and this
CorrSensOpts.Correlation.SampleSize = 1e4;
CorrAnalysis = uq_createAnalysis(CorrSensOpts);
uq_print(CorrAnalysis)
uq_display(CorrAnalysis)
```